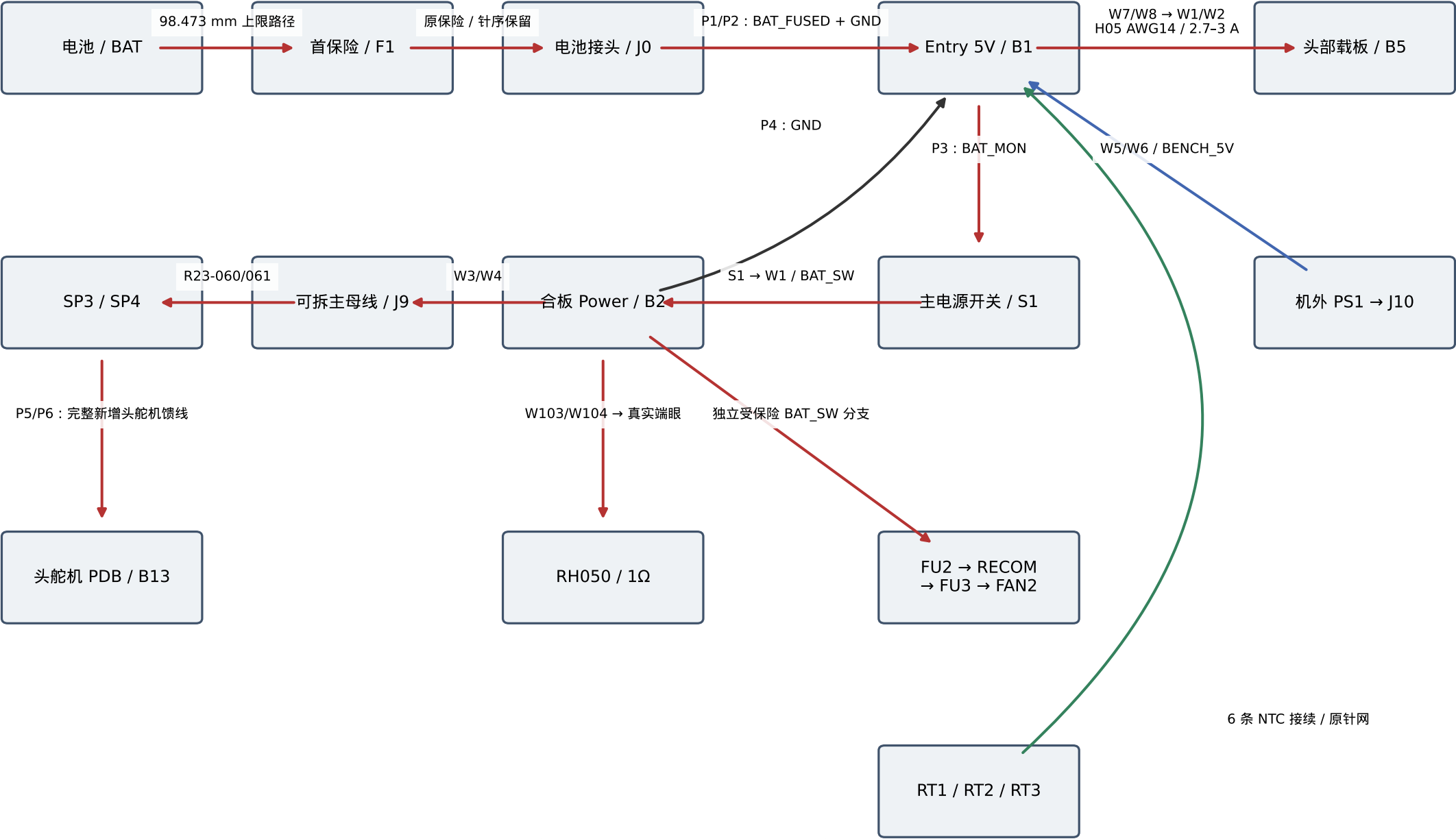


R23 电源与完整线束接线关系



跨颈：6 根功率 + 25 根信号 = 31 根。控制/温保逐针以157行净表为准；箭头仅表示端到端关系，不代表电流恒定方向。

59条新/改线路 · 逐针与剪线表 1/5

编号	起点	终点	网络	线规	装机长度 mm	毛坯 mm
R23-001	B1:J4.1	B5:J1.1	GND	28	158.942-169.942	180 ± 1
R23-002	B1:J4.2	B5:J1.2	+3V3	28	164.742-175.742	186 ± 1
R23-003	B1:J4.3	B5:J1.3	I2C_SDA	28	170.542-181.542	192 ± 1
R23-004	B1:J4.4	B5:J1.4	I2C_SCL	28	176.342-187.342	198 ± 1
R23-005	B1:J4.5	B5:J1.5	RUN_REQ	28	182.142-193.142	204 ± 1
R23-006	B1:J4.6	B5:J1.6	RUN_OK	28	187.942-198.942	209 ± 1
R23-007	B1:J4.7	B5:J1.7	TDK_IMON	28	193.742-204.742	215 ± 1
R23-008	B1:J4.8	B5:J1.8	BRAKE_ACTIVE_N	28	199.542-210.542	221 ± 1
R23-009	B1:J4.9	B5:J1.9	GND	28	205.342-216.342	227 ± 1
R23-010	B1:J4.10	B5:J1.10	GND	28	211.142-222.142	233 ± 1
R23-011	B1:J5.1	B2:J1.1	GND	28	560.991-571.991	582 ± 1
R23-012	B1:J5.2	B2:J1.2	+3V3	28	650.392-661.392	672 ± 1
R23-013	B1:J5.3	B2:J1.3	I2C_SDA	28	644.885-655.885	666 ± 1
R23-014	B1:J5.4	B2:J1.4	I2C_SCL	28	613.730-624.730	635 ± 1

完整CSV/JSON保留颜色、料号、每段归属和工艺。毛坯余量修剪弃去；JST/AMASS/Molex内部工程区间不等于厂商公差。

未改98条按原接线表保留。风扇/保险原厂引线不得与新增TE线或器件质量重复计数。

59条新/改线路 · 逐针与剪线表 2/5

编号	起点	终点	网络	线规	装机长度 mm	毛坯 mm
R23-015	B1:J5.5	B2:J1.5	RUN_REQ	28	599.844-610.844	621 ± 1
R23-016	B1:J5.6	B2:J1.6	RUN_OK	28	613.071-624.071	635 ± 1
R23-017	B1:J5.7	B2:J1.7	TDK_IMON	28	590.832-601.832	612 ± 1
R23-018	B1:J5.8	B2:J1.8	GND	28	605.049-616.049	627 ± 1
R23-019	B1:J6.1	B6:J1.1	GND	28	713.998-724.998	735 ± 1
R23-020	B1:J6.2	B6:J1.2	+3V3	28	680.494-691.494	702 ± 1
R23-021	B1:J6.3	B6:J1.3	RUN_OK	28	743.930-754.930	765 ± 1
R23-022	B1:J7.1	B7:J1.1	GND	28	734.699-745.699	756 ± 1
R23-023	B1:J7.2	B7:J1.2	+3V3	28	731.750-742.750	753 ± 1
R23-024	B1:J7.3	B7:J1.3	RUN_OK	28	735.224-746.224	757 ± 1
R23-042	B1:J8.1	B2:P101.1	GND	28	549.353-554.853	565 ± 1
R23-043	B1:J8.2	B2:P102.1	+3V3	28	609.965-615.465	626 ± 1
R23-044	B1:J8.3	B2:P103.1	I2C_SDA	28	563.834-569.334	580 ± 1
R23-045	B1:J8.4	B2:P104.1	I2C_SCL	28	587.088-592.588	603 ± 1

完整CSV/JSON保留颜色、料号、每段归属和工艺。毛坯余量修剪弃去；JST/AMASS/Molex内部工程区间不等于厂商公差。

未改98条按原接线表保留。风扇/保险原厂引线不得与新增TE线或器件质量重复计数。

59条新/改线路 · 逐针与剪线表 3/5

编号	起点	终点	网络	线规	装机长度 mm	毛坯 mm
R23-046	B1:J8.5	B2:P105.1	BRAKE_ACTIVE_N	28	569.459-574.959	585 ± 1
R23-047	B1:J1.1	RT1:1	NTC_BAT	28	768.716-774.216	785 ± 1
R23-048	B1:J1.2	RT1:2	GND	28	770.630-776.130	787 ± 1
R23-049	B1:J2.1	RT2:1	NTC_TDK	28	660.798-666.298	677 ± 1
R23-050	B1:J2.2	RT2:2	GND	28	659.833-665.333	676 ± 1
R23-051	B1:J3.1	RT3:1	NTC_BRAKE	28	682.749-688.249	699 ± 1
R23-052	B1:J3.2	RT3:2	GND	28	709.147-714.647	725 ± 1
R23-053	J0:1	B1:W1.1	BAT_FUSED	18_then16	696.624-702.624	713 ± 1
R23-054	J0:2	B1:W2.1	GND	18_then16	596.648-602.648	613 ± 1
R23-055	B1:W3.1	S1:1	BAT_MON	16	520.360	531 ± 1
R23-056	S1:3	B2:W1.1	BAT_SW	16	81.619	92 ± 1
R23-057	B1:W4.1	B2:W2.1	GND	16	536.422	547 ± 1
R23-058	B2:W3.1	J9:1	SERVO_BUS	16	132.992	143 ± 1
R23-059	B2:W4.1	J9:2	GND	16	194.699	205 ± 1

完整CSV/JSON保留颜色、料号、每段归属和工艺。毛坯余量修剪弃去；JST/AMASS/Molex内部工程区间不等于厂商公差。

未改98条按原接线表保留。风扇/保险原厂引线不得与新增TE线或器件质量重复计数。

59条新/改线路 · 逐针与剪线表 4/5

编号	起点	终点	网络	线规	装机长度 mm	毛坯 mm
R23-060	J9:1	SP3:1	SERVO_BUS	16	138.837-144.837	155 ± 1
R23-061	J9:2	SP4:1	GND	16	137.136-143.136	154 ± 1
R23-062	B2:W103.1	RH1:1	SERVO_BUS	16	155.391	166 ± 1
R23-063	B2:W104.1	RH1:2	RH_LOW	16	273.064	284 ± 1
R23-064	B1:W7.1	B5:W1.1	SYSTEM_5V	14	174.651	185 ± 1
R23-065	B1:W8.1	B5:W2.1	GND	14	177.651	188 ± 1
R23-101	SP3:3	B12:W1.1	SERVO_BUS	16	143.462	154 ± 1
R23-118	SP3:4	B13:W1.1	SERVO_BUS	16	691.869	702 ± 1
R23-119	SP4:4	B13:W2.1	GND	16	757.823	768 ± 1
R23-138	PS1:1	B1:W5.1	BENCH_5V	20	71.494-78.954	80 ± 0.5
R23-139	PS1:2	B1:W6.1	GND	20	70.884-78.344	80 ± 0.5
R23-152	B2:W1.1	FU2:1	BAT_SW	factory fuse lead	6.785	原厂引线修剪
R23-153	FU2:2	FANREG:1	FANREG_VIN	24	215.739	原厂引线修剪
R23-154	B2:W2.1	FANREG:2	GND	24	251.923	原厂引线修剪

完整CSV/JSON保留颜色、料号、每段归属和工艺。毛坯余量修剪弃去；JST/AMASS/Molex内部工程区间不等于厂商公差。

未改98条按原接线表保留。风扇/保险原厂引线不得与新增TE线或器件质量重复计数。

59条新/改线路 · 逐针与剪线表 5/5

编号	起点	终点	网络	线规	装机长度 mm	毛坯 mm
R23-155	FANREG:3	FU3:1	FAN2_5V_RAW	factory fuse lead	18.856	原厂引线修剪
R23-156	FU3:2	FAN2:1	FAN2_5V	factory fan AWG2	97.559	原厂引线修剪
R23-157	FANREG:2	FAN2:2	GND	factory fan AWG2	115.093	原厂引线修剪

完整CSV/JSON保留颜色、料号、每段归属和工艺。毛坯余量修剪弃去；JST/AMASS/Molex内部工程区间不等于厂商公差。

未改98条按原接线表保留。风扇/保险原厂引线不得与新增TE线或器件质量重复计数。

全机157行端到端净表 1/6

编号	起点	终点	系统网络	颜色	状态
R23-001	B1:j4.1	B5:j1.1	GND	black	replaced
R23-002	B1:j4.2	B5:j1.2	+3V3	red	replaced
R23-003	B1:j4.3	B5:j1.3	I2C_SDA	green	replaced
R23-004	B1:j4.4	B5:j1.4	I2C_SCL	yellow	replaced
R23-005	B1:j4.5	B5:j1.5	RUN_REQ	white	replaced
R23-006	B1:j4.6	B5:j1.6	RUN_OK	violet	replaced
R23-007	B1:j4.7	B5:j1.7	TDK_IMON	blue	replaced
R23-008	B1:j4.8	B5:j1.8	BRAKE_ACTIVE_N	brown	replaced
R23-009	B1:j4.9	B5:j1.9	GND	black	replaced
R23-010	B1:j4.10	B5:j1.10	GND	black	replaced
R23-011	B1:j5.1	B2:j1.1	GND	black	replaced
R23-012	B1:j5.2	B2:j1.2	+3V3	red	replaced
R23-013	B1:j5.3	B2:j1.3	I2C_SDA	green	replaced
R23-014	B1:j5.4	B2:j1.4	I2C_SCL	yellow	replaced
R23-015	B1:j5.5	B2:j1.5	RUN_REQ	white	replaced
R23-016	B1:j5.6	B2:j1.6	RUN_OK	violet	replaced
R23-017	B1:j5.7	B2:j1.7	TDK_IMON	blue	replaced
R23-018	B1:j5.8	B2:j1.8	GND	black	replaced
R23-019	B1:j6.1	B6:j1.1	GND	black	replaced
R23-020	B1:j6.2	B6:j1.2	+3V3	red	replaced
R23-021	B1:j6.3	B6:j1.3	RUN_OK	violet	replaced
R23-022	B1:j7.1	B7:j1.1	GND	black	replaced
R23-023	B1:j7.2	B7:j1.2	+3V3	red	replaced
R23-024	B1:j7.3	B7:j1.3	RUN_OK	violet	replaced
R23-025	B5:j2.1	B10:j1.1	GND	black	retained
R23-026	B5:j2.2	B10:j1.2	+3V3	red	retained
R23-027	B5:j2.3	B10:j1.3	I2C_SDA	green	retained
R23-028	B5:j2.4	B10:j1.4	I2C_SCL	yellow	retained

retained=原线路继承；replaced=已有完整继任路径；new=风扇新增分支。原155行去向另见映射CSV（98/53/4）。

全机157行端到端净表 2/6

编号	起点	终点	系统网络	颜色	状态
R23-029	B5:j3.1	B8:j1.1	GND	black	retained
R23-030	B5:j3.2	B8:j1.2	+3V3	red	retained
R23-031	B5:j3.3	B8:j1.3	I2S_BCLK	source R13 / labeled pin	retained
R23-032	B5:j3.4	B8:j1.4	GND	black	retained
R23-033	B5:j3.5	B8:j1.5	I2S_WS	source R13 / labeled pin	retained
R23-034	B5:j3.6	B8:j1.6	I2S_TX	source R13 / labeled pin	retained
R23-035	B5:j3.7	B8:j1.7	I2S_RX	source R13 / labeled pin	retained
R23-036	B5:j3.8	B8:j1.8	AMP_ENABLE	source R13 / labeled pin	retained
R23-037	B8:j4.1	B9:j1.1	GND	black	retained
R23-038	B8:j4.2	B9:j1.2	+3V3	red	retained
R23-039	B8:j4.3	B9:j1.3	I2S_BCLK	source R13 / labeled pin	retained
R23-040	B8:j4.4	B9:j1.4	I2S_WS	source R13 / labeled pin	retained
R23-041	B8:j4.5	B9:j1.5	I2S_RX	source R13 / labeled pin	retained
R23-042	B1:j8.1	B2:P101.1	GND	black	replaced
R23-043	B1:j8.2	B2:P102.1	+3V3	red	replaced
R23-044	B1:j8.3	B2:P103.1	I2C_SDA	green	replaced
R23-045	B1:j8.4	B2:P104.1	I2C_SCL	yellow	replaced
R23-046	B1:j8.5	B2:P105.1	BRAKE_ACTIVE_N	brown	replaced
R23-047	B1:j1.1	RT1:1	NTC_BAT	white	replaced
R23-048	B1:j1.2	RT1:2	GND	black	replaced
R23-049	B1:j2.1	RT2:1	NTC_TDK	white	replaced
R23-050	B1:j2.2	RT2:2	GND	black	replaced
R23-051	B1:j3.1	RT3:1	NTC_BRAKE	white	replaced
R23-052	B1:j3.2	RT3:2	GND	black	replaced
R23-053	J0:1	B1:W1.1	BAT_FUSED	red	replaced
R23-054	J0:2	B1:W2.1	GND	black	replaced
R23-055	B1:W3.1	S1:1	BAT_MON	red	replaced
R23-056	S1:3	B2:W1.1	BAT_SW	orange	replaced

retained=原线路继承；replaced=已有完整继任路径；new=风扇新增分支。原155行去向另见映射CSV（98/53/4）。

全机157行端到端净表 3/6

编号	起点	终点	系统网络	颜色	状态
R23-057	B1:W4.1	B2:W2.1	GND	black	replaced
R23-058	B2:W3.1	J9:1	SERVO_BUS	red	replaced
R23-059	B2:W4.1	J9:2	GND	black	replaced
R23-060	J9:1	SP3:1	SERVO_BUS	red	replaced
R23-061	J9:2	SP4:1	GND	black	replaced
R23-062	B2:W103.1	RH1:1	SERVO_BUS	red	replaced
R23-063	B2:W104.1	RH1:2	RH_LOW	blue	replaced
R23-064	B1:W7.1	B5:W1.1	SYSTEM_5V	red	replaced
R23-065	B1:W8.1	B5:W2.1	GND	black	replaced
R23-066	B5:J8.1	B8:J2.1	GND	black	retained
R23-067	B5:J8.2	B8:J2.2	AUDIO_5V	source R13 / labeled pin	retained
R23-068	B5:J7.1	FAN1:-	GND	black	retained
R23-069	B5:J7.2	FAN1:+	FAN_5V	source R13 / labeled pin	retained
R23-070	B8:J3.1	SPK1:1	SPK_P	source R13 / labeled pin	retained
R23-071	B8:J3.2	SPK1:2	SPK_N	source R13 / labeled pin	retained
R23-072	B5:J4.1	USB1:USB1	U2D2_5V	source R13 / labeled pin	retained
R23-073	B5:J5.1	CAM1:USB1	UVC_5V	source R13 / labeled pin	retained
R23-074	B5:J6.1	USB0:1	OTG_VBUS_ONLY	source R13 / labeled pin	retained
R23-075	B5:J4.2	USB1:USB2	U2D2_DM	source R13 / labeled pin	retained
R23-076	B5:J5.2	CAM1:USB2	UVC_DM	source R13 / labeled pin	retained
R23-077	B5:J6.2	USB0:2	OTG_DM	source R13 / labeled pin	retained
R23-078	B5:J4.3	USB1:USB3	U2D2_DP	source R13 / labeled pin	retained
R23-079	B5:J5.3	CAM1:USB3	UVC_DP	source R13 / labeled pin	retained
R23-080	B5:J6.3	USB0:3	OTG_DP	source R13 / labeled pin	retained
R23-081	B5:J4.4	USB1:USB5	GND	black	retained
R23-082	B5:J5.4	CAM1:USB4	GND	black	retained
R23-083	B5:J6.4	USB0:4	GND	black	retained
R23-084	SP3:2	B11:W1.1	SERVO_BUS	red	retained

retained=原线路继承；replaced=已有完整继任路径；new=风扇新增分支。原155行去向另见映射CSV（98/53/4）。

全机157行端到端净表 4/6

编号	起点	终点	系统网络	颜色	状态
R23-085	SP4:2	B11:W2.1	GND	black	retained
R23-086	B11:J1.1	M20:1	GND	black	retained
R23-087	B11:J1.2	M20:2	PDB_L_V_SERVO_1	source R13 / labeled pin	retained
R23-088	B11:J1.3	M20:3	DXL_DATA	source R13 / labeled pin	retained
R23-089	B11:J2.1	M21:1	GND	black	retained
R23-090	B11:J2.2	M21:2	PDB_L_V_SERVO_2	source R13 / labeled pin	retained
R23-091	B11:J2.3	M21:3	DXL_DATA	source R13 / labeled pin	retained
R23-092	B11:J3.1	M22:1	GND	black	retained
R23-093	B11:J3.2	M22:2	PDB_L_V_SERVO_3	source R13 / labeled pin	retained
R23-094	B11:J3.3	M22:3	DXL_DATA	source R13 / labeled pin	retained
R23-095	B11:J4.1	M23:1	GND	black	retained
R23-096	B11:J4.2	M23:2	PDB_L_V_SERVO_4	source R13 / labeled pin	retained
R23-097	B11:J4.3	M23:3	DXL_DATA	source R13 / labeled pin	retained
R23-098	B11:J5.1	M24:1	GND	black	retained
R23-099	B11:J5.2	M24:2	PDB_L_V_SERVO_5	source R13 / labeled pin	retained
R23-100	B11:J5.3	M24:3	DXL_DATA	source R13 / labeled pin	retained
R23-101	SP3:3	B12:W1.1	SERVO_BUS	red	replaced
R23-102	SP4:3	B12:W2.1	GND	black	retained
R23-103	B12:J1.1	M10:1	GND	black	retained
R23-104	B12:J1.2	M10:2	PDB_R_V_SERVO_1	source R13 / labeled pin	retained
R23-105	B12:J1.3	M10:3	DXL_DATA	source R13 / labeled pin	retained
R23-106	B12:J2.1	M11:1	GND	black	retained
R23-107	B12:J2.2	M11:2	PDB_R_V_SERVO_2	source R13 / labeled pin	retained
R23-108	B12:J2.3	M11:3	DXL_DATA	source R13 / labeled pin	retained
R23-109	B12:J3.1	M12:1	GND	black	retained
R23-110	B12:J3.2	M12:2	PDB_R_V_SERVO_3	source R13 / labeled pin	retained
R23-111	B12:J3.3	M12:3	DXL_DATA	source R13 / labeled pin	retained
R23-112	B12:J4.1	M13:1	GND	black	retained

retained=原线路继承；replaced=已有完整继任路径；new=风扇新增分支。原155行去向另见映射CSV（98/53/4）。

全机157行端到端净表 5/6

编号	起点	终点	系统网络	颜色	状态
R23-113	B12:J4.2	M13:2	PDB_R_V_SERVO_4	source R13 / labeled pin	retained
R23-114	B12:J4.3	M13:3	DXL_DATA	source R13 / labeled pin	retained
R23-115	B12:J5.1	M14:1	GND	black	retained
R23-116	B12:J5.2	M14:2	PDB_R_V_SERVO_5	source R13 / labeled pin	retained
R23-117	B12:J5.3	M14:3	DXL_DATA	source R13 / labeled pin	retained
R23-118	SP3:4	B13:W1.1	SERVO_BUS	red	replaced
R23-119	SP4:4	B13:W2.1	GND	black	replaced
R23-120	B13:J1.1	M30:1	GND	black	retained
R23-121	B13:J1.2	M30:2	PDB_H_V_SERVO_1	source R13 / labeled pin	retained
R23-122	B13:J1.3	M30:3	DXL_DATA	source R13 / labeled pin	retained
R23-123	B13:J2.1	M31:1	GND	black	retained
R23-124	B13:J2.2	M31:2	PDB_H_V_SERVO_2	source R13 / labeled pin	retained
R23-125	B13:J2.3	M31:3	DXL_DATA	source R13 / labeled pin	retained
R23-126	B13:J3.1	M32:1	GND	black	retained
R23-127	B13:J3.2	M32:2	PDB_H_V_SERVO_3	source R13 / labeled pin	retained
R23-128	B13:J3.3	M32:3	DXL_DATA	source R13 / labeled pin	retained
R23-129	M32:1	M33:1	GND	black	retained
R23-130	M32:2	M33:2	PDB_H_V_SERVO_3	source R13 / labeled pin	retained
R23-131	M32:3	M33:3	DXL_DATA	source R13 / labeled pin	retained
R23-132	M33:1	M34:1	GND	black	retained
R23-133	M33:2	M34:2	PDB_H_V_SERVO_3	source R13 / labeled pin	retained
R23-134	M33:3	M34:3	DXL_DATA	source R13 / labeled pin	retained
R23-135	USB1:TTL3	B11:J6.2	DXL_DATA	source R13 / labeled pin	retained
R23-136	B11:J6.4	B12:J6.2	DXL_DATA	source R13 / labeled pin	retained
R23-137	B12:J6.4	B13:J6.2	DXL_DATA	source R13 / labeled pin	retained
R23-138	PS1:1	B1:W5.1	BENCH_5V	red	replaced
R23-139	PS1:2	B1:W6.1	GND	black	replaced
R23-140	FU1:W2.1	J0:1	BAT_FUSED	red	retained

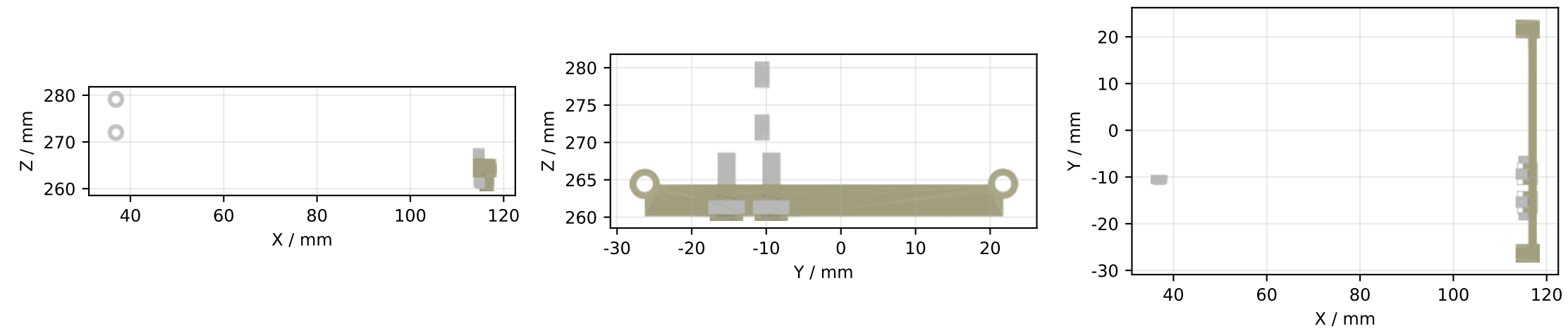
retained=原线路继承；replaced=已有完整继任路径；new=风扇新增分支。原155行去向另见映射CSV（98/53/4）。

全机157行端到端净表 6/6

编号	起点	终点	系统网络	颜色	状态
R23-141	PCM1:P-	J0:2	GND	black	retained
R23-142	PCM1:B+	FU1:W1.1	CELL4	source R13 / labeled pin	retained
R23-143	CELL ASSEMBLY:NODE0	PCM1:B-	CELL0	source R13 / labeled pin	retained
R23-144	CELL ASSEMBLY:NODE1	PCM1:B1	CELL1	source R13 / labeled pin	retained
R23-145	CELL ASSEMBLY:NODE2	PCM1:B2	CELL2	source R13 / labeled pin	retained
R23-146	CELL ASSEMBLY:NODE3	PCM1:B3	CELL3	source R13 / labeled pin	retained
R23-147	CELL ASSEMBLY:NODE4	PCM1:B+	CELL4	source R13 / labeled pin	retained
R23-148	J0:1	JC1:1	CHARGE_POS	source R13 / labeled pin	retained
R23-149	J0:2	JC1:2	CHARGE_RETURN	source R13 / labeled pin	retained
R23-150	PS2:1	JC1:1	CHARGE_16V8	source R13 / labeled pin	retained
R23-151	PS2:2	JC1:2	CHARGE_GND	source R13 / labeled pin	retained
R23-152	B2:W1.1	FU2:1	BAT_SW	red sleeve	new
R23-153	FU2:2	FANREG:1	FANREG_VIN	red	new
R23-154	B2:W2.1	FANREG:2	GND	black	new
R23-155	FANREG:3	FU3:1	FAN2_5V_RAW	red sleeve	new
R23-156	FU3:2	FAN2:1	FAN2_5V	red	new
R23-157	FANREG:2	FAN2:2	GND	black	new

retained=原线路继承；replaced=已有完整继任路径；new=风扇新增分支。原155行去向另见映射CSV（98/53/4）。

H05 B5 真实端接与末端桥 · 原生实体正投影



PEEK桥复用后隔柱1/3；原孔轴与5 mm叠层保留。两条线各3.6 mm夹持，R2裸铜为固定弯。

真实全长174.651391566 /177.651391566 mm；3 mm搭焊，0.05 mm名义焊料底层。未实测。

冻结范围与首件验收

- 354个唯一安装对象；29层具名来源；729个来源/几何摘要绑定；188个原生端点逐针核对。
- 59条完整新/改线、157行最终净表、155行旧去向（98保留、53替换、4删除）。
- H05 3 A条件最低4.759441495 V，仅9.44 mV余量；2.7 A为4.776393756 V。原PCB最坏条件仍适用。
- P5/P6全线热态电阻30.519546 mΩ；RH两线9.020025 mΩ，需联合原生开关/分流/热模型。
- J10工程热态整回路 ≤ 25 mΩ；3 A服务侧加载 ≥ 5.025 V、空载 ≤ 5.20 V，固定标称5.0 V不自动满足。
- FU2/FU3的300 A为分断能力；无电池/BMS最坏故障电流保证时，短路配合尚未证实。
- 头部31根服务回环为有限姿态几何候选；根集成另核9773连续R21区间。实际恢复力、疲劳和热漂移未测。
- 先做台架端接/系带再装壳；背夹/J10/NTC各自装配顺序不可互换。J9维护先断电、断J0、取出电池。
- 质量依照Complete_Conductor_Lengths与Conductor_Mass_Allocation，只按旧具名段扣旧加新；58视觉预算不扣铜重。
- 所有加工公差、供应商端子内部尺寸、焊料/压接/绝缘/拉力、热态风量/漏气均按子目录验收，未作制造放行。